



Yakeen Plus (2025)

Short Practice Test - 08

DURATION: 60 Minutes

DATE: 10/11/2024

M. MARKS: 192

Topics Covered

Physics:	Electromagnetic Waves, Ray Optics and Optical Instruments, Wave Optics, Alternating Current
Chemistry:	Practical Organic Chemistry, Coordination Compound, The d and f-Block Elements, Biomolecules
Biology:	(Botany) : Principles of inheritance and variations, microbes in human welfare, molecular basis of inheritance (Zoology) : Human Health and Diseases, Reproductive Health

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

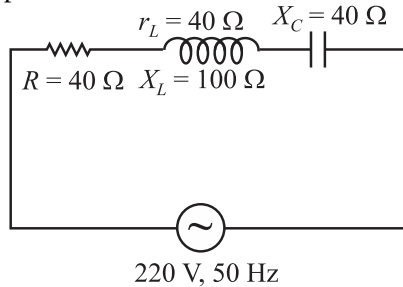
1. An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t)$; $B = B_0 \sin(kx - \omega t)$. Which of the following equations is true?

(1) $E_0 k = B_0 \omega$ (2) $E_0 \omega = B_0 k$
(3) $E_0 B_0 = \omega k$ (4) None of these

2. The quality factor of series L - C - R circuit is Q , then the value of capacitance in terms of resistance R and inductance (L) is

(1) $\frac{L^2}{R^2 Q^2}$ (2) $C = \frac{L}{(RQ)^2}$
(3) $\frac{R^2 Q^2}{L^2}$ (4) $\frac{(RQ)^2}{L}$

3. The power factor of the circuit shown in figure is



(1) 0.2 (2) 0.4
(3) 0.8 (4) 0.6

4. In Young's double slit experiment, 62 fringes are seen in visible region for sodium light of wavelength $5893 \text{ \AA}$. If violet light of wavelength $4358 \text{ \AA}$ is used in place of sodium light, then number of fringes seen will be:

(1) 54 (2) 64
(3) 74 (4) 84

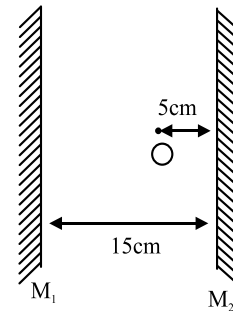
5. If in Fraunhofer diffraction due to a single slit, the slit width is increased, the width of the central maximum will

(1) increase
(2) decrease
(3) not change
(4) change depending upon the wave length of light used

6. Unpolarized light of intensity I passes through an ideal polarizer A . Another identical polarizer B is placed behind A . The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B . The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is:

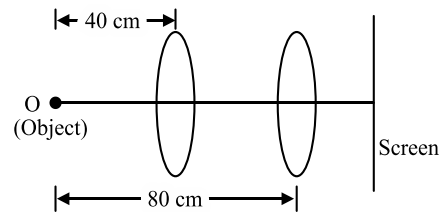
(1) 45° (2) 60°
(3) 0° (4) 30°

7. Figure shows two plane mirrors parallel to each other and an object O placed between them. Then the distance of the first three images from the mirror M_2 will be: (in cm)



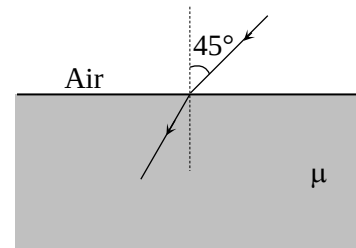
(1) 5, 10, 15
(2) 5, 15, 30
(3) 5, 25, 35
(4) 5, 15, 25

8. In displacement method, there are two positions of a lens for which we get clear image. First position of the lens is at 40 cm from object and second is at 80 cm, the focal length of lens is:



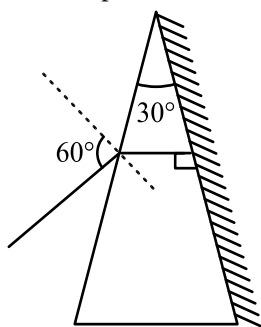
(1) 40 cm (2) $\frac{40}{3} \text{ cm}$
(3) 80 cm (4) $\frac{80}{3} \text{ cm}$

9. In the figure shown, for an angle of incidence 45° , at the top surface, what is the minimum refractive index needed for total internal reflection at vertical face?



(1) $\frac{\sqrt{2}+1}{2}$
(2) $\sqrt{\frac{3}{2}}$
(3) $\sqrt{\frac{1}{2}}$
(4) $\sqrt{2}+1$

10. An isosceles prism of angle $A = 30^\circ$ has one of its surfaces silvered. Light rays falling at an angle of incidence 60° on the other surface retrace their path after reflection from the silvered surface. The refractive index of prism material is:



- (1) 1.414
(2) 1.5
(3) 1.732
(4) 1.866

11. If electric field in an electromagnetic wave is given by $E = 6 \times 10^6 \sin\left(2\pi ft - \frac{2\pi}{\lambda} x\right) \frac{V}{m}$. The peak value of magnetic field in the region is
- (1) 20 T (2) 2 T
(3) 0.2 T (4) 0.02 T

12. According to modified Ampere's circuital law (i_D = displacement current)

- (1) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C + \epsilon_0 \frac{d\phi_E}{dt} \right)$
(2) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$
(3) $\oint \vec{B} \cdot d\vec{l} = \mu_0 i$
(4) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C \frac{d\phi_E}{dt} + i_D \right)$

[Short Practice Test-08 | Yakeen Plus (2025) | 10/11/2024]

Practical Organic Chemistry, Coordination Compound, The d and f-Block Elements, Biomolecules

SECTION-(II) CHEMISTRY

13. Glucose $\xrightarrow{Br_2 + H_2O}$ Product is:

- (1) Glucaric acid
(2) Gluconic acid
(3) Hexanoic acid
(4) Bromohexane

14. The base present in RNA but not found in DNA is:

- (1) Thymine
(2) Adenine
(3) Uracil
(4) Guanine

15. Which of the following is a non-reducing sugar?

- (1) Glucose
(2) Sucrose
(3) Maltose
(4) Lactose

16. Which of the following **cannot** distinguish among primary, secondary and tertiary alcoholic compounds?

- (1) Lucas test
(2) Victor meyer test
(3) Ceric ammonium nitrate test
(4) Reaction with sodium metal.

17. In 3rd group cationic analysis, NH_4Cl is added before adding NH_4OH . This is due to:

- (1) Decrease the concentration of OH^-
(2) Increase the concentration of OH^-
(3) Increase concentration of Cl^-
(4) Prevent interference of PO_4^{3-} ion

18. Which of the following is not a double salt but is a complex salt?

- (1) $KCl.MgCl_2.6H_2O$
(2) $FeSO_4.(NH_4)_2SO_4.6H_2O$
(3) $K_2SO_4.Al_2(SO_4)_3.24H_2O$
(4) $4KCN.Fe(CN)_2$

19. Hybridisation of $[Ni(NH_3)_6]^{2+}$ and $[Zn(NH_3)_6]^{2+}$ are:

- (1) Both d^2sp^3
(2) Both sp^3d^2
(3) d^2sp^3 and sp^3d^2 respectively
(4) sp^3d^2 and d^2sp^3 respectively

20. Give the IUPAC name of the complex compound $[Co(NH_3)_4(H_2O)Br](NO_3)_2$:

- (1) Bromidoaquotetraaminocobalt(III) nitrate
(2) Bromidoaquotetraaminocobalt(III) nitrate
(3) Bromidoaquatetraamminecobalt(III) nitrate
(4) TetraammineaquaBromidocobalt(III) nitrate

21. In which of the following oxidation state of Mn is six?

- (1) K_2MnO_4 (2) MnO_2
(3) $KMnO_4$ (4) MnO_4^-

22. What will be the most common oxidation state shown by the Lanthanoid?

- (1) -3 (2) +3
(3) -4 (4) +4

23. Which one of the following pairs of ions have same electronic configuration?

- (1) Cr^{3+} , Fe^{3+}
- (2) Mn^{2+} , Fe^{3+}
- (3) Fe^{3+} , Co^{3+}
- (4) Sc^{3+} , Cr^{3+}

24. Which of the following element has maximum composition in misch metal?

- (1) La
- (2) Fe
- (3) Ce
- (4) Lu

[Short Practice Test-08 | Yakeen Plus (2025) | 10/11/2024]

Principles of inheritance and variations, microbes in human welfare, molecular basis of inheritance

SECTION-(III) BOTANY

25. Select the **incorrect** statement w.r.t. DNA fingerprinting.

- (1) DNA from a single cell is enough to perform DNA fingerprinting analysis.
- (2) DNA fingerprinting has much wider applications in determining population and genetic diversities.
- (3) The VNTR belongs to a class of non-repetitive DNA.
- (4) DNA fingerprint differs from individual to individual in a population except in the case of monozygotic twins.

26. Identify the microorganism which is responsible for the production of an immunosuppressive molecule cyclosporin A:

- (1) *Trichoderma polysporum*.
- (2) *Clostridium butylicum*.
- (3) *Aspergillus niger*.
- (4) *Saccharomyces cerevisiae*.

27. The promoter site and terminator site of transcription unit are located at:

- (1) 3' (downstream) and 5' (upstream) end respectively of the coding strand.
- (2) 5' - end (upstream) of the structural gene and 3' - end (downstream) of the coding strand.
- (3) 5' (upstream) end of the coding strand.
- (4) 3' (downstream) end of the structural gene.

28. Which of the following can be used as a biocontrol agent of several plant pathogen?

- (1) *Azospirillum*
- (2) *Lactobacillus*
- (3) *Trichoderma*
- (4) *Azotobacter*

29. Select the **correct** statements from the following w.r.t transcription.

- A. The DNA-dependent DNA polymerase catalyse the polymerisation in only one direction.
- B. The coding sequences or expressed sequences are defined as introns.
- C. Intervening sequences do appear in processed RNA.
- D. In bacteria, translation can begin much before the mRNA is fully transcribed.

Choose the most appropriate answer from the option given below:

- (1) A and D (2) A and C
- (3) B and C (4) B and D

30. Broad palm with characteristic palm crease is visible in a person suffering from:

- (1) Turner's syndrome.
- (2) Klinefelter's syndrome.
- (3) Thalassema.
- (4) Down's syndrome.

31. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In Griffith's experiment, a mixture of heat-killed virulent bacteria S and live non-virulent bacteria R, lead to the death of mice.

Reason R: 'Transforming principle' got transferred from heat killed S strain to live R strain and made it virulent.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

32. In a sewage treatment plant, primary treatment is:

- (1) a physical process which involves sedimentation only.
- (2) a physical process which involves both filtration and sedimentation.
- (3) a biological process which formation of primary sludge and effluent.
- (4) a biological process which involves both filtration and sedimentation.

33. In sickle cell anaemia:
- (1) the mutant haemoglobin molecule undergoes polymerisation under low oxygen tension causing the change in the shape of RBC.
 - (2) substitution of glutamic acid by valine at the sixth position of the α – chain of haemoglobin.
 - (3) the mutant haemoglobin undergoes polymerization under high oxygen tension causing the change in shape of RBC.
 - (4) α -globin chain is modified.

34. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Aspergillus niger</i>	(I)	Acetic Acid
(B)	<i>Acetobacter aceti</i>	(II)	Lactic Acid
(C)	<i>Clostridium butylicum</i>	(III)	Citric Acid
(D)	<i>Lactobacillus</i>	(IV)	Butyric Acid

Choose the **correct** answer from the options given below.

- (1) A-IV, B-II, C-I, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-III, C-I, D-IV

35. What is/are **true** about BOD?

- A. BOD is directly proportional to polluting potential.
 - B. The BOD test measures the rate of uptake of nitrogen by microorganisms.
 - C. BOD is the indirect measure of inorganic matter present in water.
- (1) Only A
 - (2) Only A and B
 - (3) Only B and C
 - (4) All A, B and C

36. Match **List-I** with **List-II**.

List-I		List-II	
(A)	XX-XO method of sex Determination	(I)	Klinefelter's Syndrome
(B)	XX-XY method of sex Determination	(II)	Female heterogametic
(C)	Karyotype-47	(III)	Grasshopper
(D)	ZW-ZZ method of sex Determination	(IV)	Female homogametic

Choose the **correct** answer from the options given below.

- (1) A-II, B-IV, C-I, D-III
- (2) A-I, B-IV, C-II, D-III
- (3) A-III, B-IV, C-I, D-II
- (4) A-IV, B-II, C-I, D-III

[Short Practice Test-08 | Yakeen Plus (2025) | 10/11/2024]
Human Health and Diseases, Reproductive Health

SECTION-(IV) ZOOLOGY

37. Which of the following statements is **incorrect**?
- (1) A widely used diagnostic test for AIDS is ELISA.
 - (2) Genital herpes is a curable STDs.
 - (3) Macrophages act as HIV factory.
 - (4) Saheli is a non-steroidal pill.
38. Identify the diagram given below and select the **correct** option describing its function.



- (1) It activates immune system and helps in destroying the tumor.
- (2) It is a depressant and slows down body functions.
- (3) It is used as pain inducer.
- (4) All of these

39. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
- Assertion A:** In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the foetal cells and dissolved substances.
- Reason R:** Amniocentesis is used legally to check the gender of developing foetus.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R true and R is the correct explanation of A.
 - (4) Both A and R true and R is not the correct explanation of A.

40. Which of the following options is **incorrectly** matched?

(1)	IUI	Semen collected from husband or donor is artificially introduced into the uterus
(2)	GIFT	Transfer of embryos with more than 8 blastomeres into the fallopian tube
(3)	ICSI	Sperm are directly injected into the ovum
(4)	ZIFT	Transfer of embryos upto 8 blastomeres into the fallopian tube

41. How many of the following diseases is/are sexually transmitted diseases(STDs)?

Syphilis, chlamydia, hepatitis-B, genital warts, HIV
--

- (1) One (2) Three
(3) Two (4) Five

42. Given below are two statements.

Statement I: Rejection of a kidney graft is the function of B-lymphocytes.

Statement II: The antibodies against small pox pathogen are produced by T - lymphocytes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
(2) Statement I is incorrect, but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

43. All of the following statements are true for an IUD **except**:

- (1) It is one of the most widely accepted methods of contraception for males in India.
(2) Progestasert is a hormone releasing IUDs.
(3) IUD is an ideal contraceptive to delay pregnancies.
(4) IUD increases phagocytosis of sperms within the uterus.

44. Identify the **incorrect** match related to the causes, detection, and diagnosis of cancer:

(1)	Carcinogens	Agents that can transform normal cells into cancerous neoplastic cells
(2)	Ionizing radiations	Include X-rays and gamma rays, cause DNA damage leading to neoplastic transformation
(3)	Oncogenic viruses	Viruses with genes called viral oncogenes, leading to cancer
(4)	MRI	Uses strong magnetic fields and ionising radiations to detect changes in living tissue

45. The site where lymphocytes interact with antigens and proliferate to become effector cells are:

- (1) spleen and lymph nodes.
(2) bone marrow and thymus.
(3) Peyer's patches and tonsils.
(4) both (1) and (3)

46. Choose the **correct** option w.r.t the same kind of immunity barrier.

- (1) Polymorphonuclear leucocytes and monocyte
(2) Skin and tears in eyes
(3) Saliva in mouth and interferons
(4) Mucus lining of urinogenital tract and HCl in stomach

47. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Coitus interruptus	(I)	Block gamete transport
(B)	Cu7	(II)	Withdrawing penis before ejaculation
(C)	Vaults	(III)	Reduces the motility and fertilising capacity of sperms
(D)	Vasectomy	(IV)	Prevent physical meeting of sperm and ovum

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D- II
(2) A-II, B-III, C-IV, D- I
(3) A-III, B-I, C-IV, D- II
(4) A- II, B-I, C-IV, D-III

48. Read the following statements (A-E) w.r.t typhoid fever.

- A. *Wuchereria bancrofti* is a causative agent of typhoid fever in human beings.
- B. It is caused by consuming contaminated food and water.
- C. In severe cases dry scaly lesions appear on skin and scalp.

- D. Sustained high fever, weakness, stomach pain and loss of appetite are some symptoms of it.
- E. Pathogens of typhoid fever usually enters the large intestine.

How many of the above statements is/are **correct**?

- (1) One
- (2) Two
- (3) Four
- (4) Five



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